The effects of Immigration Enforcement on Health Care Utilization: A National Analysis

Naomi Buell   Richie Rivera

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1 Abstract (to be completed for final report)

Background: The removal of "sensitive location" protections and increased ICE activity function can discourage people from seeking medical care by creating fear, stress, and other barriers to accessing health services, including among immigrant communities not directly targeted by enforcement.

Objective: To assess whether increased ICE enforcement correlates with reduced health care utilization.

Methods: We use national datasets on ICE detention activity and health care utilization to analyze the relationship between enforcement intensity and health care utilization across geographies and over time. We will control for potential confounders such as socioeconomic status, demographic characteristics, and local health care infrastructure.

Results: [TBD]

Conclusions: We found that immigration enforcement [is/is not] associated with reduced health care utilization. [Implications for public health and policy TBD]

2 Introduction

Immigration-related fears appear to be negatively affecting health care access in the United States (Kaiser Family Foundation 2025). Recent survey evidence indicates that nearly half (48%) of likely undocumented immigrant adults and 14% of immigrant adults overall report that they or a family member have avoided seeking medical care due to immigration-related concerns. These findings suggest that immigration enforcement may function as a structural barrier to health care access, even for individuals who are not directly targeted by enforcement actions. In January 2025, federal policy changes reversed prior protections that designated hospitals and other health care facilities as "sensitive locations," limiting enforcement activity in these settings (Department of Homeland Security 2025). The removal of these protections may plausibly increase fears associated with seeking care, particularly in communities with visible or intensified U.S. Immigration and Customs Enforcement (ICE) activity.

Motivated by these developments, this project investigates the relationship between immigration enforcement intensity and health care utilization. Specifically, we seek to answer the question: *Does increased ICE enforcement correlate with reduced health care utilization?* We hypothesize that higher levels of local enforcement activity are associated with greater health care avoidance, as individuals may be less willing to leave their homes or engage with formal institutions perceived as potential sites of immigration enforcement.

2.1 Literature Review

Empirical research provides strong evidence that immigration enforcement activity reduces health care utilization among immigrant populations. At the state level, Hispanic patients were significantly less likely to report having a regular provider or annual checkup following increased ICE activity, a pattern not observed among non-Hispanic adults (Friedman and Venkataramani 2021). This differential response underscores how enforcement intensity functions as a racialized barrier to care. A more dramatic illustration comes from research on Secure Communities, an immigration enforcement program expanded during the Obama administration from 2008 to 2014, which found that likely authorized Hispanic immigrants experienced a 16.9% decline in the probability of visiting a health care provider compared to non-Hispanic U.S.-born respondents (Herring and Barnow 2025). Complementing these state-level findings, a survey of Asian and Latinx immigrants in California demonstrated that each additional direct experience with immigration enforcement was associated with delayed care for both groups (Young et al. 2023).

Beyond individual-level utilization changes, enforcement has broader structural impacts on health care access. Studies using geographic measures of enforcement intensity, such as ICE detainer requests at the county level, found these requests were associated with lower Medicaid and SNAP enrollment—indicating that enforcement creates barriers to accessing public benefits and formal health care systems (Kravitz et al. 2024). At the regional level, Latino adults living in areas with the most detainer requests had 24% higher odds of fair/poor self-rated health compared to those in areas with the least requests, reflecting enforcement's effects on health outcomes through reduced care access (Eastus et al. 2025). These findings align with a comprehensive literature review synthesizing 32 studies, which documented that hostile immigrant environments correlate with decreased or delayed utilization of basic health care services (Vernice et al. 2020).

The mechanisms linking enforcement to reduced care-seeking are increasingly well-understood through provider and qualitative research. Surveys of clinicians consistently show enforcement's disruptive effects: 48% of primary care and emergency medicine providers reported observing negative effects of ICE enforcement on their immigrant patients' health and health access (Hacker et al. 2012), while another survey found that post-2016 anti-immigrant policies and enforcement actions led half of clinicians to report persistent fear among their immigrant patients—an effect more pronounced in border states (Saadi et al. 2021). Qualitative investigations reveal the emotional and structural pathways through which enforcement affects care-seeking. A 2009 study in Everett, Massachusetts found that enhanced enforcement impacted immigrant health through both emotional means (deportation fears, concerns about ICE-police cooperation) and structural barriers (anxiety about documentation requirements for enrollment and care) (Hacker et al. 2011). An immigration raid in a Midwestern Latino community similarly demonstrated that enforcement actions increase immigration-related stress and lower self-rated health among affected residents (Lopez et al. 2017).

While prior studies have well-documented enforcement effects through state-level analyses, provider surveys, and local case studies, comprehensive research examining the relationship on a national scale at granular geographic and temporal levels remains limited. This project addresses this gap by using national datasets on ICE detention activity and health care utilization to analyze the enforcement-utilization relationship across geographies and over time, allowing us to quantify a relationship that prior research has established in principle but not comprehensively measured at scale.

3 Methods

See Appendix A for detailed Python code used in data processing and analysis.

3.1 Data

We utilize data on immigration enforcement, health care utilization data, and social determinants of health. Detention Facilities Average Daily Population data is available from the Transactional Records Access Clearinghouse (TRAC) and provides information on the average daily population of ICE detention facilities, broken down by city, state, ZIP code, and date from September 2019 through present (Transactional Records Access Clearinghouse 2026). This dataset will allow us to measure the intensity of local immigration enforcement activity over time and across geographies.

Routine checkup within the past year among adults data is available from CDC's PLACES data and provides information on the percentage of adults who reported having a routine checkup within the past year, broken down by county, ZCTA, and year from 2020-2025 (Centers for Disease Control and Prevention 2025). This dataset will serve as our primary outcome variable, allowing us to assess health care utilization patterns in relation to local enforcement intensity.

We will use data from the American Community Survey (ACS) 5-Year Estimates to control for social determinants of health in our analysis (U.S. Census Bureau 2018-2023). This data

Table 1: Descriptive statistics

Variable	Mean	SD
Percent of adults who had a visit with a doctor or other health care professional in the past year	76	4.7
Percent of population in ICE detention facilities	0.00014	0.0084
Hispanic/Latino	0.097	0.16
White	0.81	0.22
Black	0.077	0.16
Asian	0.023	0.059
Two or more races	0.044	0.058
Other race	0.046	0.11
Non-citizen	0.031	0.055
Household income	62740	32635
Below poverty line	0.13	0.11
Unemployed	0.054	0.061
High school education or higher	0.89	0.096
Without internet access	0.16	0.13
Without a vehicle	0.062	0.094

provides information on social determinants of health at the ZCTA level. Social determinants of health analyzed included race, citizenship status, income, education, employment status, and housing conditions.

Our panel data spans from 2018 to 2023 and includes 32,977 unique ZIP Code Tabulation Areas (ZCTAs) across the United States. Table 1 provides descriptive statistics for our key variables of interest, including the average daily population in ICE detention facilities, the percentage of adults reporting a routine checkup within the past year, and key social determinants of health across ZCTAs.

Figure 1 provides a visual representation of the correlations between a subset of our analysis variables. The population in ICE detention facilities does not seem to be strongly correlated with the rate of doctor visits in a particular area. Hispanic/Latino ethnicity and citizenship status are much stronger correlates with doctor visits. Figure 2 further shows that there is no identifiable linear correlation between the two variables. ZCTAs with disparate levels of healthcare utilization are distributed relatively evenly across the entire spectrum of detention intensity. This indicates that the operational size of a detention facility, as quantified by Average Daily Population, is likely not a primary driver for the frequency with which a community accesses routine medical care.

3.2 Geographic Distribution of Health Care Utilization

The following map visualizes the percentage of adults who reported a doctor visit within the past year at the ZCTA level for 2023.

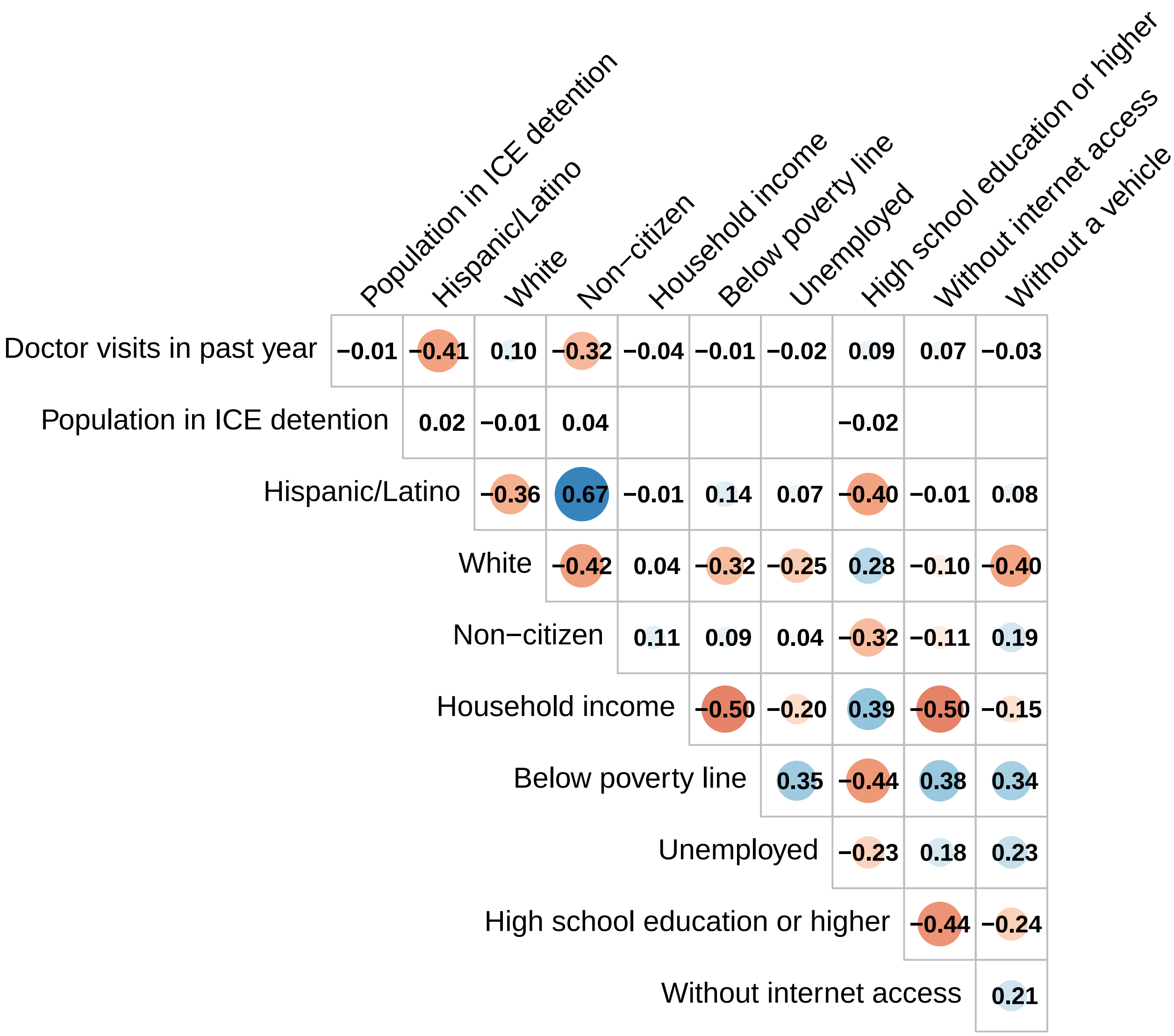

Figure 1: Correlation matrix. Note: Correlations below 0.01 are suppressed.

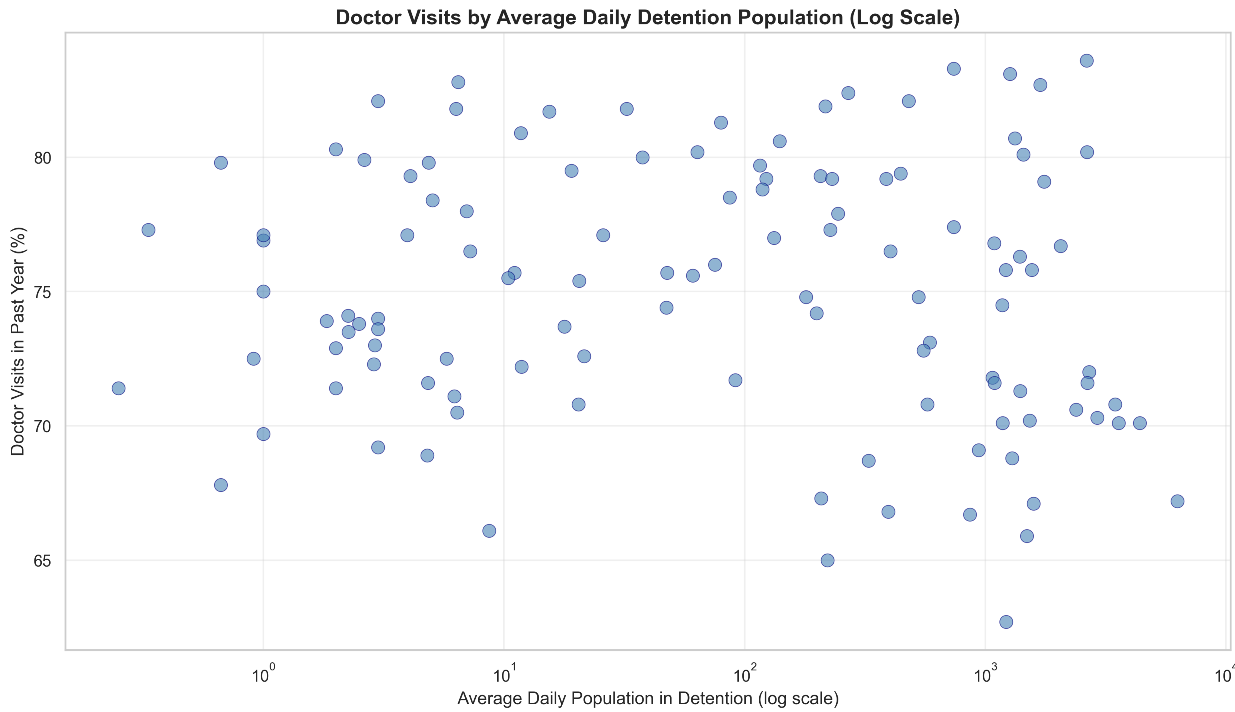

Figure 2: Scatter plot of log ICE Detentions vs. Doctor Visits

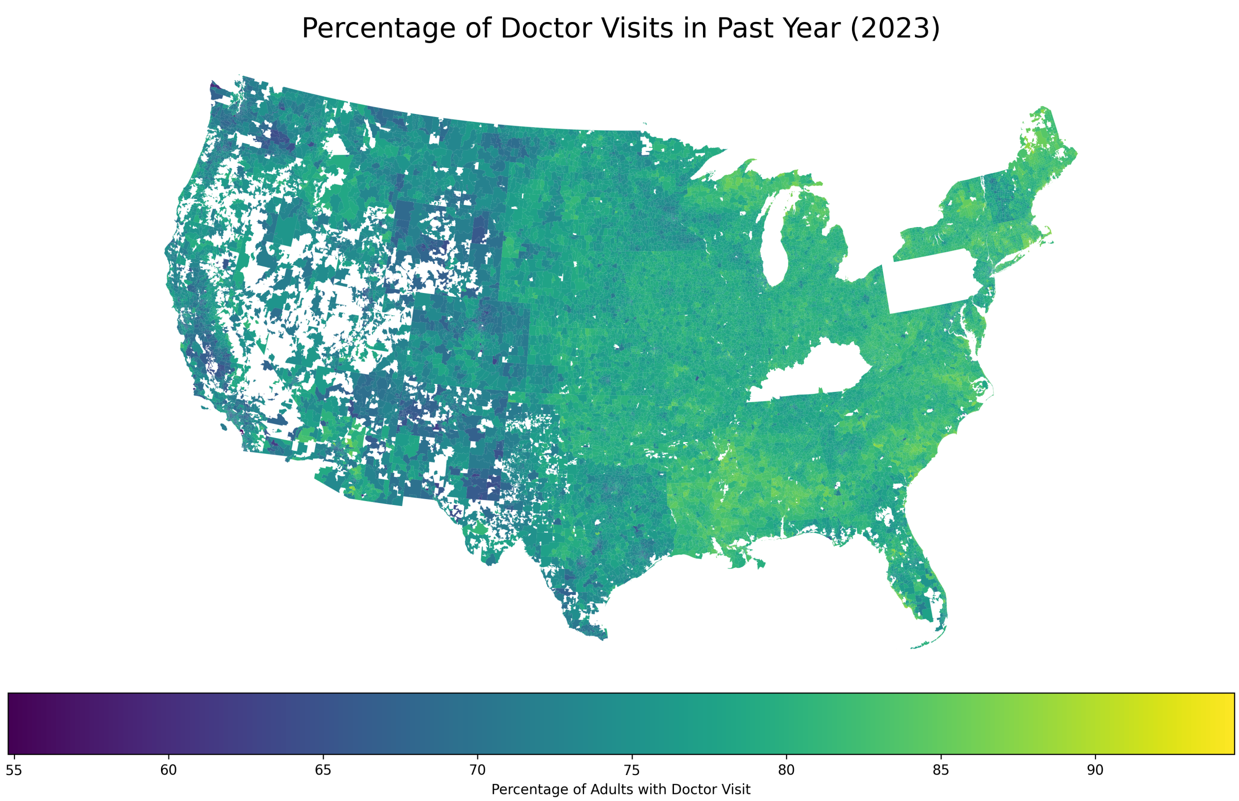

Figure 3: Map of Doctor Visits by ZCTA

The Map in Figure 3 reveals a stark divide between the national norm and the specific "cold spots" that align with our research hypothesis. The median utilization rate of 78.6% suggests that for the majority of the United States, annual healthcare engagement is a standard behavior. The tight interquartile range (75.7% to 80.6%) indicates that in most ZCTAs, about four out of five adults see a professional yearly. This represents the "health norm" from which the outliers deviate.

Spatial analysis of these outliers reveals significant geographic clustering. While the Northeast and Upper Midwest maintain high utilization, a distinct belt of low healthcare engagement is visible across the Southwest. These regions exhibit rates significantly below the national interquartile range, often falling between 55% and 65%.

3.3 Geographic Distribution of Immigration Enforcement Intensity

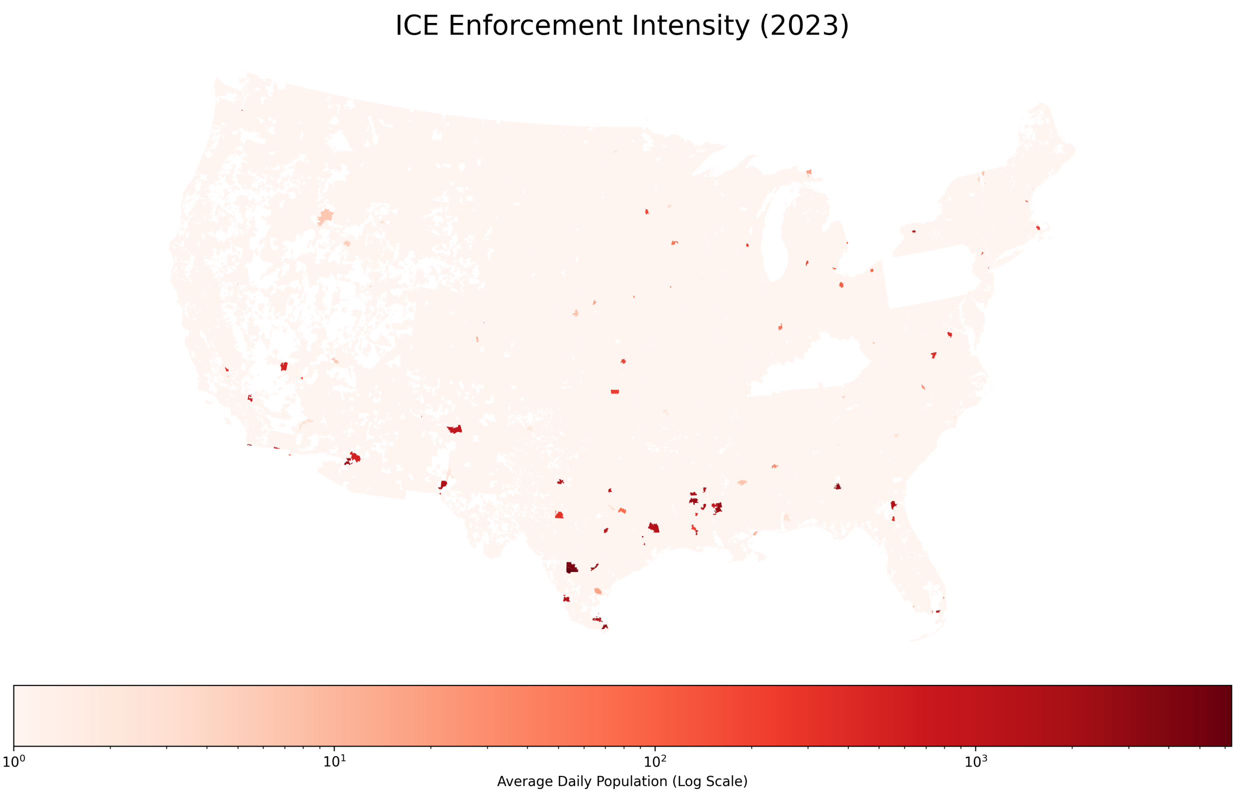

Figure 4: Map of ICE Detentions by ZCTA

The spatial distribution of immigration enforcement shown in Figure 4 illustrates the concentrated nature of federal enforcement infrastructure. Unlike general socioeconomic variables that tend to vary gradually across geographies, ICE detentions are highly localized, with a small number of ZCTAs exhibiting extremely high average daily populations (ADP) in detention facilities. The median ADP across ZCTAs is 0, indicating what we can see in the map which is that most areas have minimal enforcement activity. However, the 90th percentile ADP is 0, and the maximum ADP reaches 6281, demonstrating that certain ZCTAs experience disproportionately intense enforcement.

3.4 Doctor Visits by Citizen Status

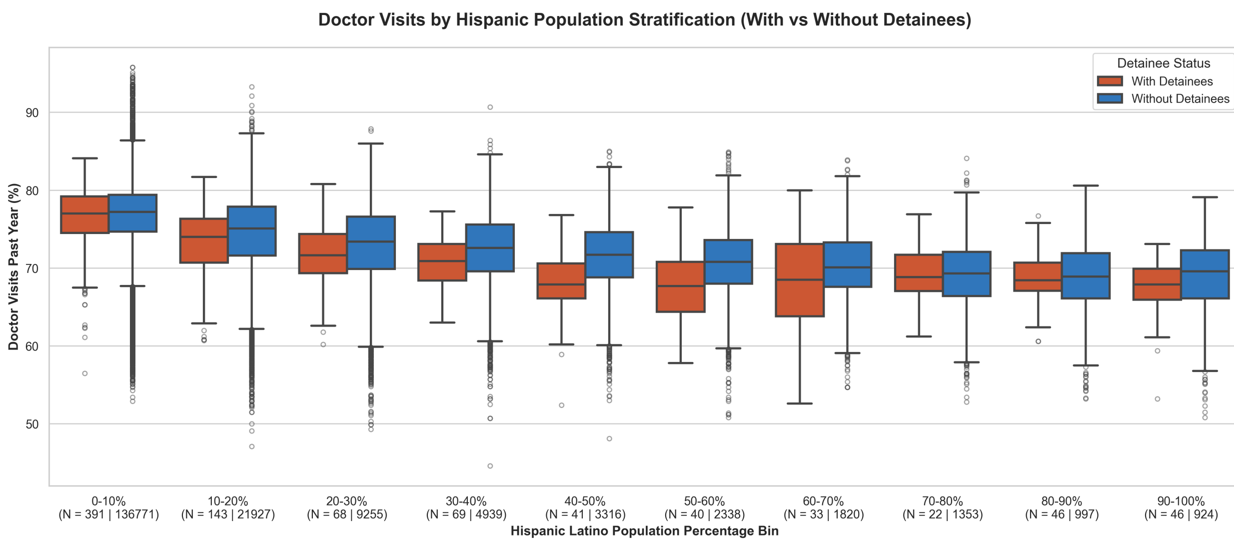

Figure 5: Box plot of doctor visits by citizenship status where there detainees

The box plot in Figure 5 shows how health care utilization varies by citizenship status across ZCTAs with detention facilities. By comparing the citizen population to the non-citizen population and differentiating between ZCTAs with and without detainees we can see how enforcement may be impacting health care utilization for different subpopulations. It is pretty clear that the median utilization rate decreases as the non-citizen population increases and this pattern is further exacerbated in ZCTAs with detention facilities. This can suggests that the presence of a detention facility may be contributing to reduced health care utilization among non-citizen populations although we cannot rule out that there may be other confounders which may be driving this relationship.

3.5 Annual Distribution of Detainee Data

The box plot in Figure 6 shows the distribution of average daily population in detention facilities across ZCTAs for each year from 2019 to 2023. The plot reveals that while the median ADP remains relatively stable across years, there was a slight reduction form 2019 through 2021 but has had a sharp increase in 2022 and 2023. The interquartile range also

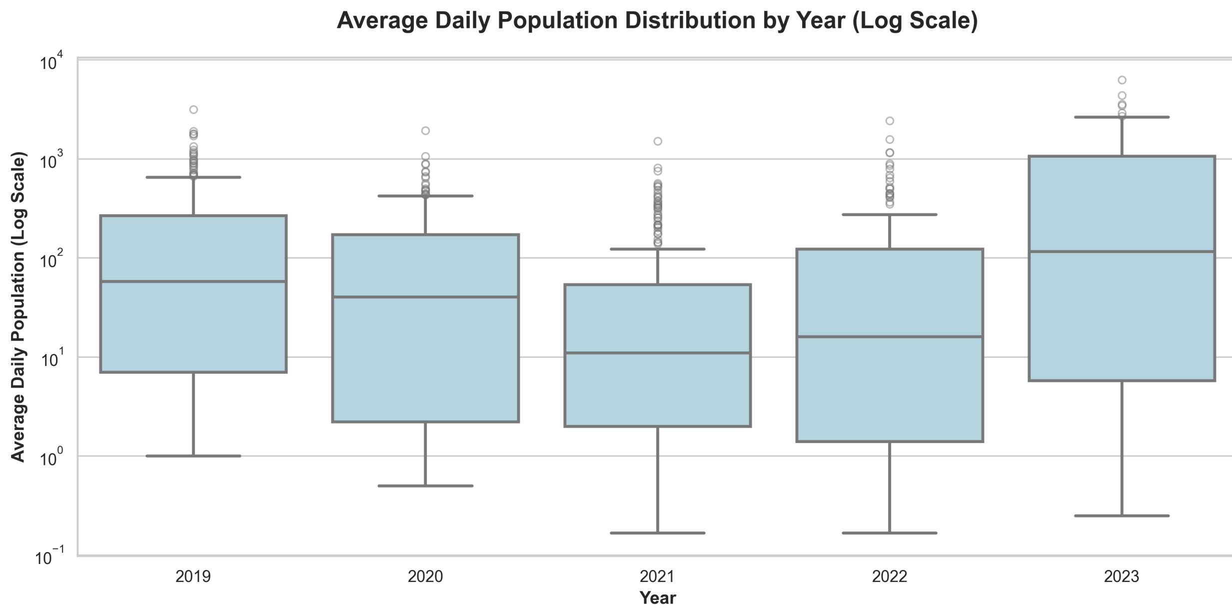

Figure 6: Box plot of average daily population in detention facilities by year

appears to widen in the later years, suggesting that while many ZCTAs continue to have low ADP, a growing number of ZCTAs are experiencing higher levels of enforcement activity.

3.6 Demographic differences in healthcare utilization

The correlation matrix in Figure 7 above shows the observed correlation between healthcare utilization and demographic compositions of ZCTAs. As we can see, there is a positive correlation between healthcare utilization and the percentage of white and black population. There is a strong negative correlation between healthcare utilization and the percentage of Hispanic/Latino and other populations with a weaker negative correlation with the Asian population. This suggests that ZCTAs with higher proportions of Hispanic/Latino have a propensity for lower healthcare utilization, which may be driven by a variety of factors.

3.7 Educational attainment and healthcare utilization

Figure 8 and Figure 9 show the relationship between healthcare utilization and educational attainment. Both plots show the observed correlation between the percentage of the population with a high school education or higher and the percentage of adults who had a doctor visit in the past year. The correlation appears to be much stronger for the high school education variable than for the bachelor's degree variable, suggesting that basic educational attainment is much more important of a factor in the population's healthcare utilization than higher education. The regression plot in Figure 10 further illustrates this relationship by showing

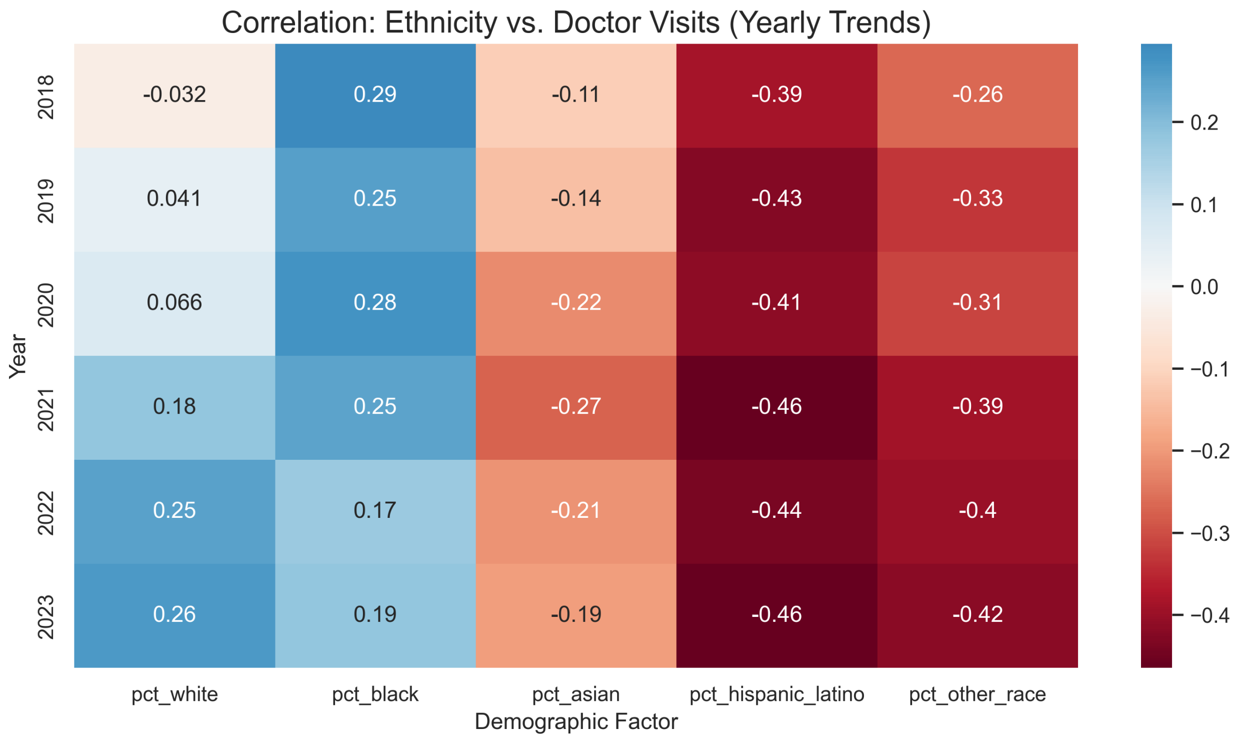

Figure 7: Correlation Matrix of doctor visits by year and ethnicity

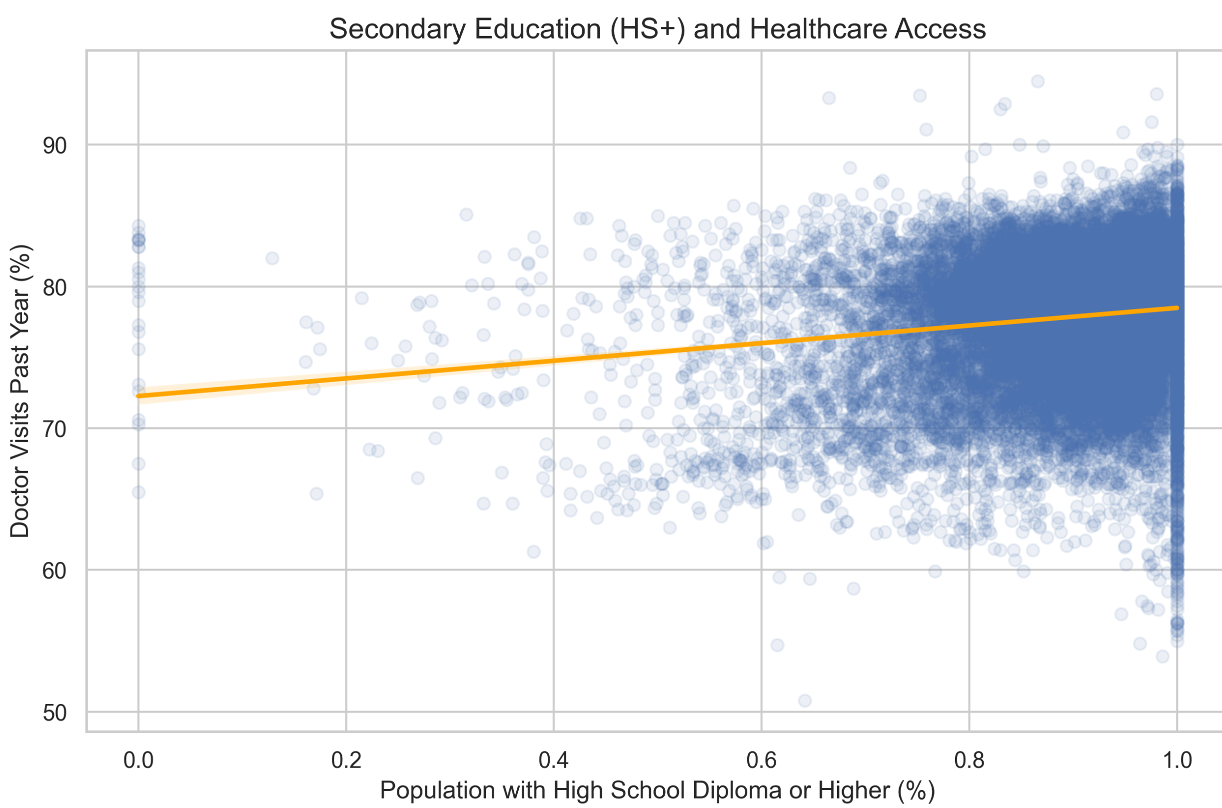

Figure 8: Scatter Plot of Doctor Visits by Education Level - Highschool or Higher

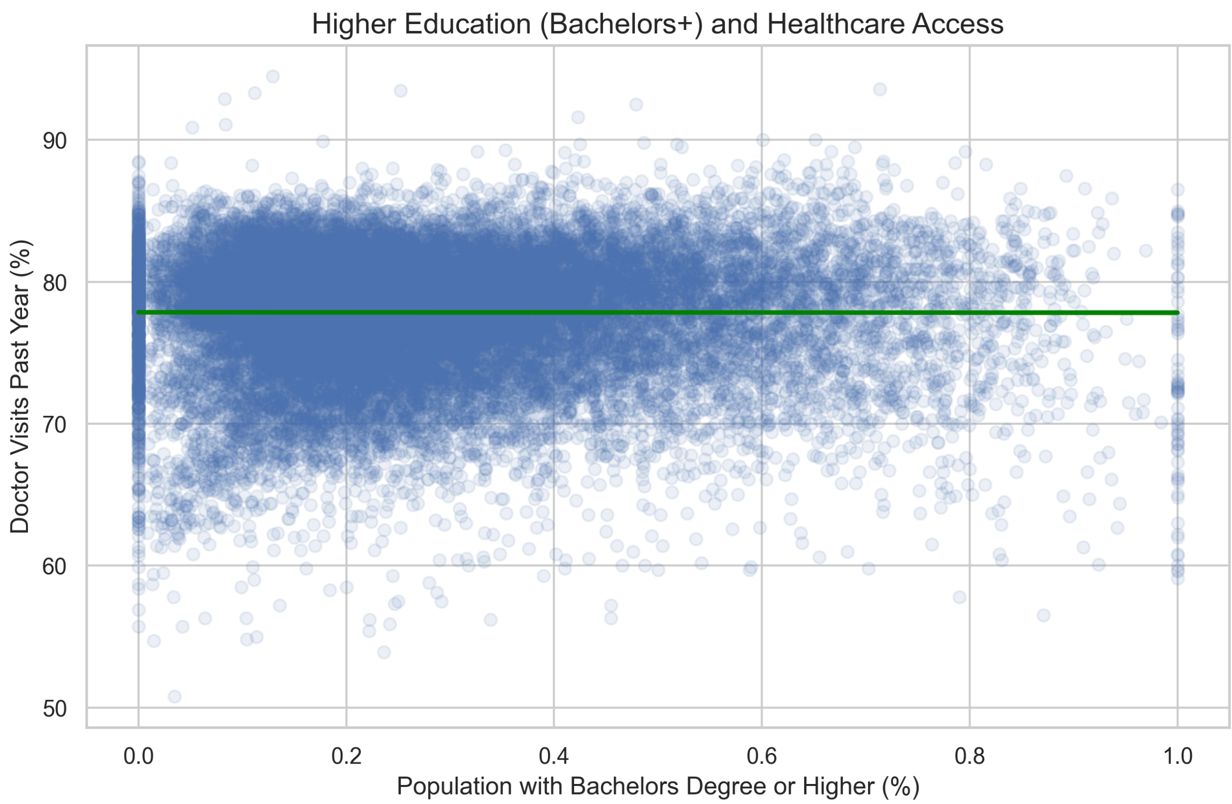

Figure 9: Scatter Plot of Doctor Visits by Education Level - Bachelors or Higher

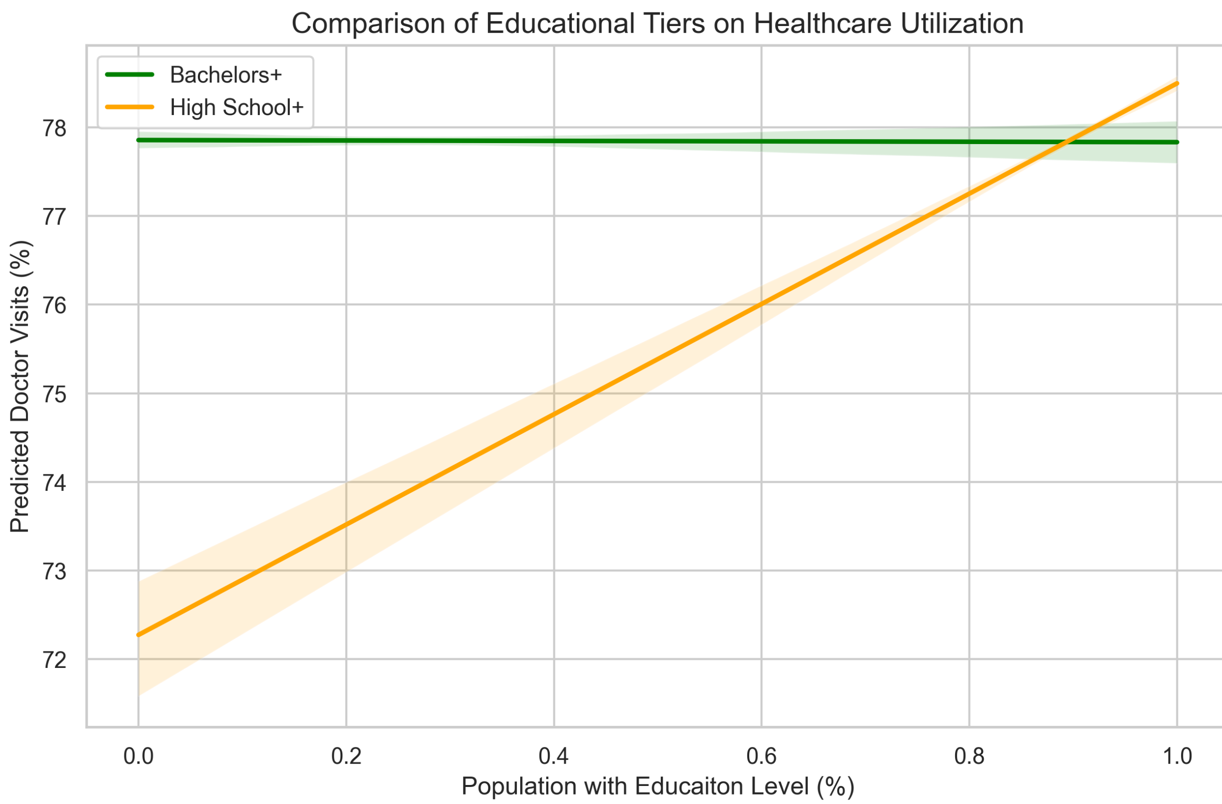

Figure 10: Regression Plot of the correlation between doctor visits education rate

the fitted regression line for the correlation between doctor visits and high school education rate, which shows a clear and strong positive relationship with high school completion rates and almost no relationship with bachelor's degree rates.

3.8 Economic Drivers and healthcare utilization

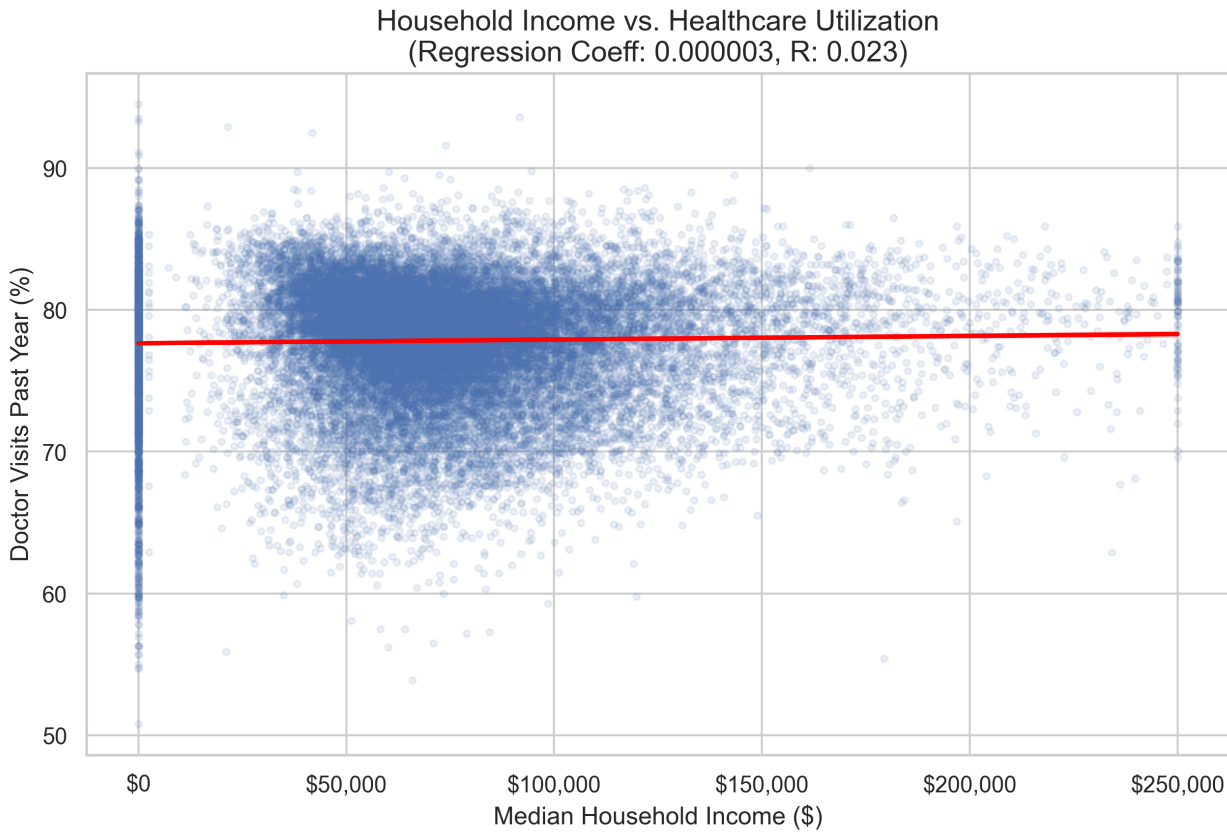

Figure 11: Scatter Plot of Doctor Visits by Median Household Income

The scatter plot in Figure 11 shows the relationship between healthcare utilization and median household income. The plot reveals that there is a very small positive correlation.

In contrast, the scatter plot in Figure 12 shows a much stronger negative correlation between healthcare utilization and unemployment rates. This suggests that economic factors, particularly unemployment, may be more significant drivers of healthcare utilization than income levels.

3.9 Analysis (in progress)

We plan to use Python to build panel data across datasets, geographies, and years. We will then use modeling techniques to analyze the relationship between immigration enforcement

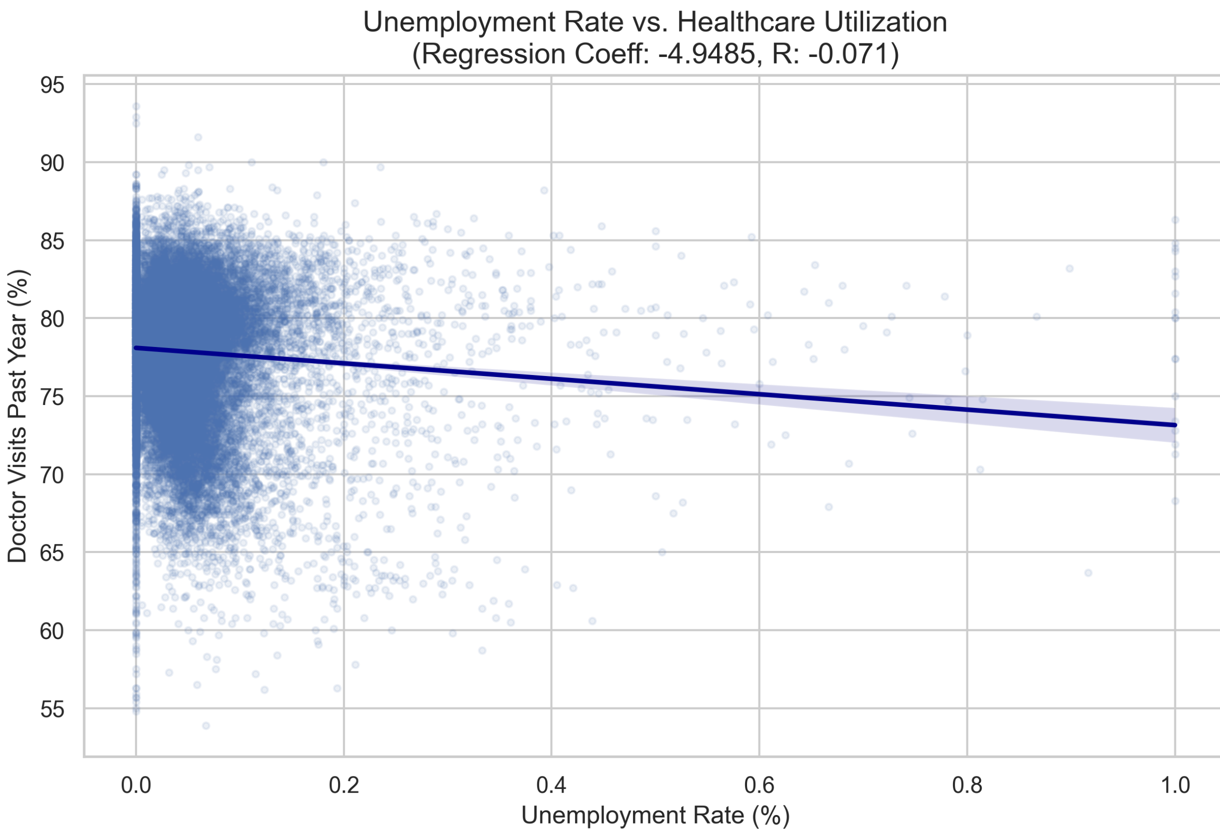

Figure 12: Scatter Plot of Doctor Visits by Unemployment Rate

intensity and health care utilization. We will control for potential confounding factors such as socioeconomic status, demographic characteristics, and local health care infrastructure. We will also explore potential heterogeneity in the relationship by subpopulations, such as by citizenship status or facility type. A potential visualization of our analysis could be a series of maps showing the geographic distribution of enforcement intensity and health care utilization, as well as scatter plots or regression lines illustrating the relationship between these variables.

4 Results (in progress)

- Results will identify specific geographic areas in need of health outreach.
- Findings highlight the hidden public health costs of immigration enforcement.
- The study provides evidence for maintaining healthcare facilities as safe zones.
- Research helps clinicians understand why immigrant patients may miss preventive care.

5 Discussion (in progress)

6 Conclusion (in progress)

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